

Janarthanan S

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SUMMARY

B.E. Electronics and Communication Engineering graduate with hands-on experience in technical troubleshooting, debugging, and root cause analysis through internship and real-world projects. Proficient in Python, REST APIs, SQL, and Git, with strong problem-solving, documentation, and communication skills. Experienced in diagnosing software and system issues, analyzing logs, and delivering reliable technical solutions. Eager to investigate complex technical problems, build automation tools, and deliver exceptional customer engineering experiences at the intersection of technology and product.

TECHNICAL SKILLS

Languages: Python, SQL, HTML, CSS

Systems & Tools: Git, GitHub, Google Cloud, Firebase, VS Code, REST APIs

Support Skills: Debugging, Troubleshooting, Issue Diagnosis, Documentation, Technical Communication

Concepts: OOP, DSA, Networking Fundamentals, Software Deployment

ML & Data: Pandas, NumPy, OpenCV, NLP, Scikit-learn

INTERNSHIP EXPERIENCE

Proxenix Software Company

July 2025 – September 2025

Software Engineer Intern – Technical Systems & ML (Remote)

- Identified and diagnosed system failures in an NLP pipeline; analyzed root causes, debugged issues, and implemented reliable solutions delivering high-accuracy results to clients.
- Monitored end-to-end software pipeline performance – data ingestion, processing, model deployment – identifying bottlenecks and resolving issues to maintain system reliability and on-time delivery.
- Documented technical findings clearly and presented solutions to cross-functional stakeholders in agile sprints, improving team alignment and communication across the project.

PROJECTS

Pothole Detection & Real-Time Alert System | Python, YOLOv11, OpenCV, System Debugging

- Identified a high false positive rate degrading system reliability; analyzed model performance, applied deep learning optimization techniques including hyperparameter tuning and dataset augmentation, and validated fixes under real-world conditions – achieving 92% precision.
- Recognized gaps in hazard detection reliability caused by single-sensor limitations; integrated radar sensor fusion with the vision pipeline to cross-verify detections, reducing missed alerts and improving overall system robustness.
- Managed the full system lifecycle from development through deployment and monitoring; implemented automated voice alerting and continuous performance checks, ensuring consistent, fault-tolerant operation in production.

Smart Hostel Issue Tracking System | Python, SQL, REST API

- Identified delays in student grievance handling caused by manual, untracked complaint logging; designed and deployed a centralized system to log, categorize, and route issues in real time – reducing resolution time significantly through structured workflows.
- Noticed recurring infrastructure complaints going unaddressed due to lack of visibility; built an admin dashboard with REST API-backed reporting, surfacing patterns that enabled proactive, data-driven decisions for administrators.

Aljust Spoken English Platform | Python, NLP, Firebase, Cloud Systems

- Identified a lack of personalized, low-latency English learning tools for regional users; built and maintained a cloud-based platform using NLP-driven voice agents and a Firebase real-time database – improving user engagement by 40%.
- Troubleshooted intermittent latency issues in real-time content delivery by profiling database queries and optimizing data sync logic, ensuring a smoother, more reliable user experience.

EDUCATION

Madras Institute of Technology, Anna University

B.E. Electronics and Communication Engineering – CGPA: 7.27/10

Sep 2023 – May 2026

Chennai, Tamil Nadu

Rajagopal Polytechnic College

Diploma in Electronics and Communication Engineering – 94.30%

2020 – 2023

Vellore, Tamil Nadu

ACHIEVEMENTS & CERTIFICATIONS

Solved 500+ DSA problems in Python, demonstrating strong analytical thinking, debugging, and problem-solving abilities.

Grew LinkedIn community to 2,000+ followers, writing about Python and technical topics for a general audience.

HackerRank Certifications: Python (Basic) | SQL (Basic) | Problem Solving (Basic) | REST API (Intermediate)

Python for Beginners – Scaler Academy | Basics of Python – Udemy

LEADERSHIP & COLLABORATION

Led cross-functional teams of 4–6 members; managed task allocation, sprint planning, and on-time delivery using agile methodology.

Mentored peers in Python and problem-solving, explaining technical concepts clearly to varied skill levels.

Completed NSO activities demonstrating discipline, adaptability, and teamwork under structured environments.